

Photography problems — Solution

X26 (2026) — English translation

A1

2.00 pts

Find the position and shape of the wire if its length is minimal.

The image of the wire is symmetrical about Y , so the wire lies in the xy plane. Note that each point of the wire gives a point on the screen (if the camera is focused on this point) or a circle (from the likeness of cones).

Let's find the position of the circle and its dimensions for the point with coordinates $(x, y, 0)$. Coordinates of the center of the circle from the similarity of triangles:

$$(Y, Z) = \left(\frac{yl}{x}, 0 \right).$$

For diameter:

$$\frac{d}{D} = \frac{a-l}{a} = 1 - \frac{l}{a}.$$

Substitute value $1/a$ from the thin lens formula.

$$\frac{d}{D} = 1 - l \left(\frac{1}{f} - \frac{1}{x} \right) = 1 - \frac{l(x-f)}{fx}.$$

Find dependence d from Y for $Y > 0$.

By condition $\tan \theta = k$.

$$d = 2Y \sin \theta = \frac{2k}{\sqrt{1+k^2}} \cdot Y = \gamma Y = \gamma \cdot \frac{yl}{x},$$

where $\gamma = \frac{2k}{\sqrt{1+k^2}}$. Substituting this into the previously obtained expression for d .

$$\frac{d}{D} = \frac{\gamma yl}{Dx} = 1 - \frac{l(x-f)}{fx}$$

Express $y(x)$:

$$y = \frac{Dx}{\gamma l} - \frac{D(x-f)}{\gamma f} = \frac{D}{\gamma} - \frac{Dx}{\gamma} \left(\frac{1}{f} - \frac{1}{l} \right) = \frac{D}{\gamma} - \frac{Dx(l-f)}{\gamma fl}.$$

Note, that since wire place in region $x > 0$ and image point wire in center screen be point, to camera focus on one from point wire, therefore $l > f$. Note, that we consider case $a \geq l$, if this not thus, to still equation remain correct, but sign y d swap. That is for consideration $a < l$ sufficiently in obtain expression change sign γ . Thus we obtain, that possible position point determine form image:

$$y = \pm \frac{D(l-f)\sqrt{1+k^2}}{2klf} \left(\frac{lf}{l-f} - x \right).$$

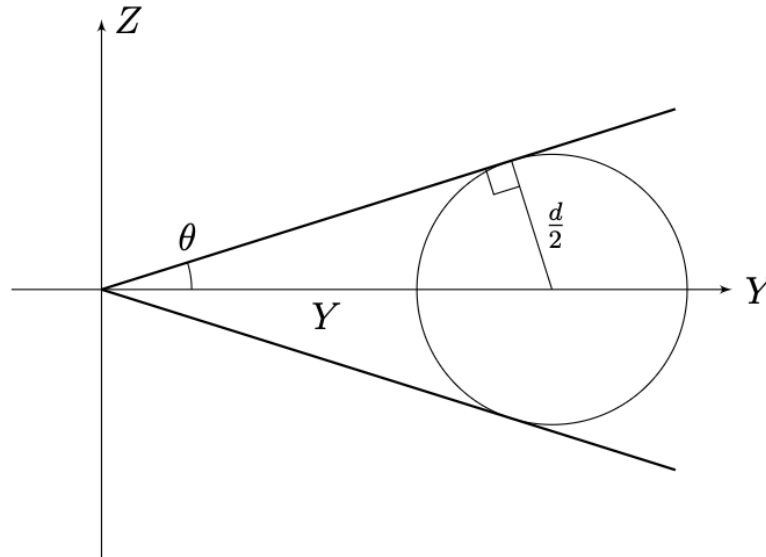
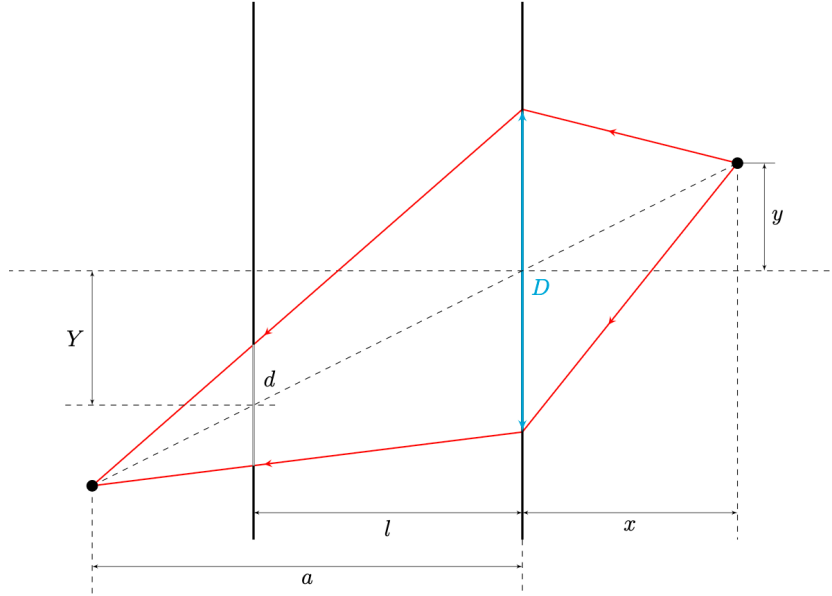
Perhaps the wire contains more sections, the images of which overlap with the images of points on these lines, but then the wire will have a greater length, which contradicts the condition. Therefore, it lies completely on these straight lines.

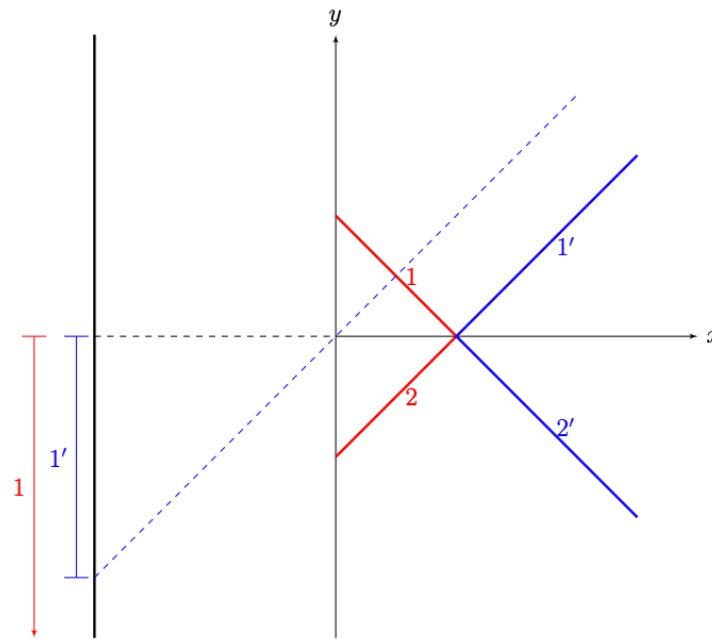
Note that the angular size of the parts of straight lines 1' and 2' is less than 90° , therefore they can give in the image a circle with a limited segment of the positions of the centers; even for a fixed angular size, the length of the wire in sections 1 and 2 will always be less, therefore the wire has the shape of an angle from the segments 1 and 2:

Answer: Answer:

$$y = \pm \frac{D(l-f)\sqrt{1+k^2}}{2klf} \left(\frac{lf}{l-f} - x \right)$$

for $x \in \left[0, \frac{lf}{l-f} \right]$.





B1

2.00 pts

Using a compass and straight edge, draw a straight line parallel to the direction of the puck's speed.

Let's understand how an image is formed. Let's enter the coordinates XY , which are attached to the puck and indicate the position of the point on it, and xy , which determine the position in the photograph. Let the speed of the line that the camera films at a particular moment is equal to u ; the center of the puck was photographed at point (x_0, y_0) and at time t_0 .

The angle between the speed of the puck and the x axis will be denoted by α . Let's find the coordinates in the photograph for an arbitrary point on the puck (X, Y) . The moment of photographing this point:

$$t = t_0 + \frac{Y}{u - v \sin \alpha}.$$

Coordinate equal:

$$x = x_0 + X + v \cos \alpha (t - t_0) = x_0 + X + \frac{v \cos \alpha}{u - v \sin \alpha} Y,$$

$$y = y_0 + u (t - t_0) = y_0 + \frac{u}{u - v \sin \alpha} Y.$$

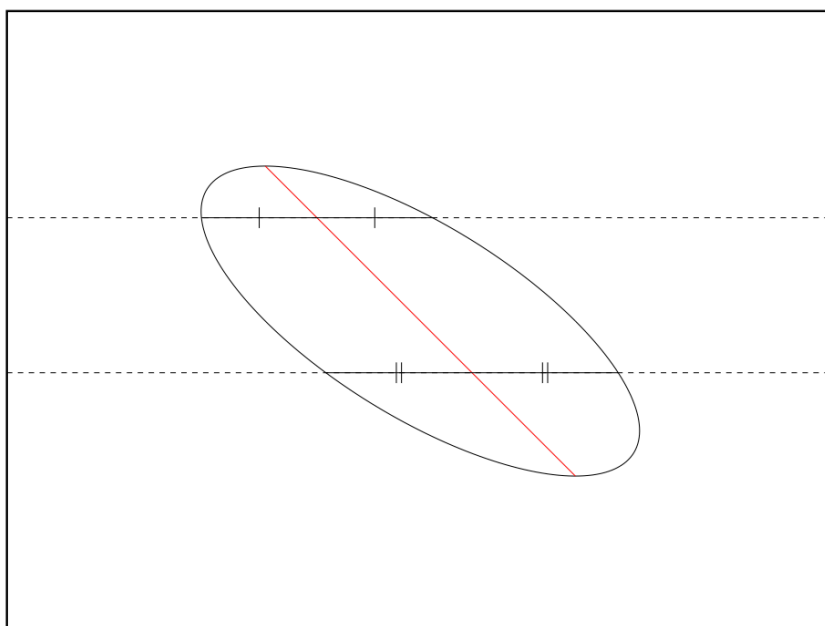
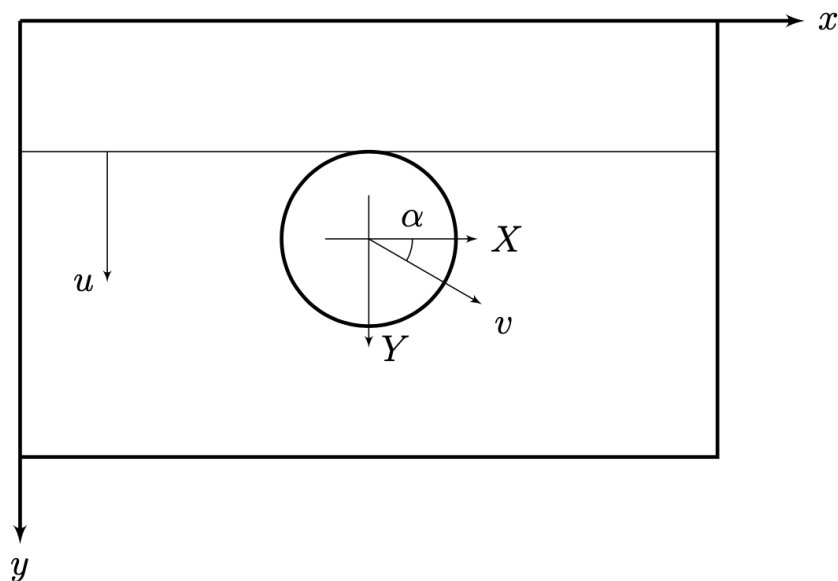
Note, that transformation coordinate linearly. Means, straight on puck be straight on image. Construct possible direction velocity. Obtain image straight $X = 0$, for this conduct horizontal straight, image $X = 0$ be straight pass through through center intersection horizontal straight with image puck (since for record Y and correspondingly y coordinate x and X differ only by a constant).

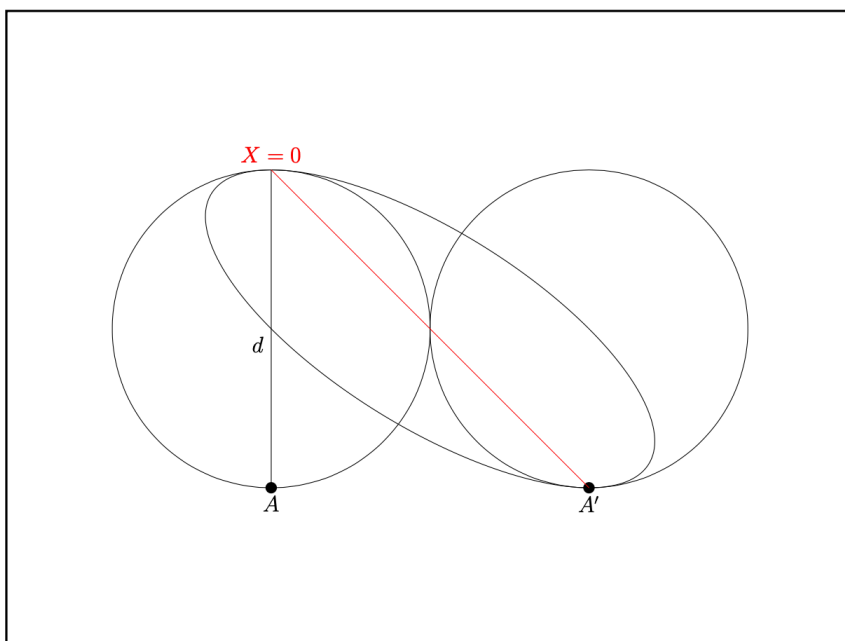
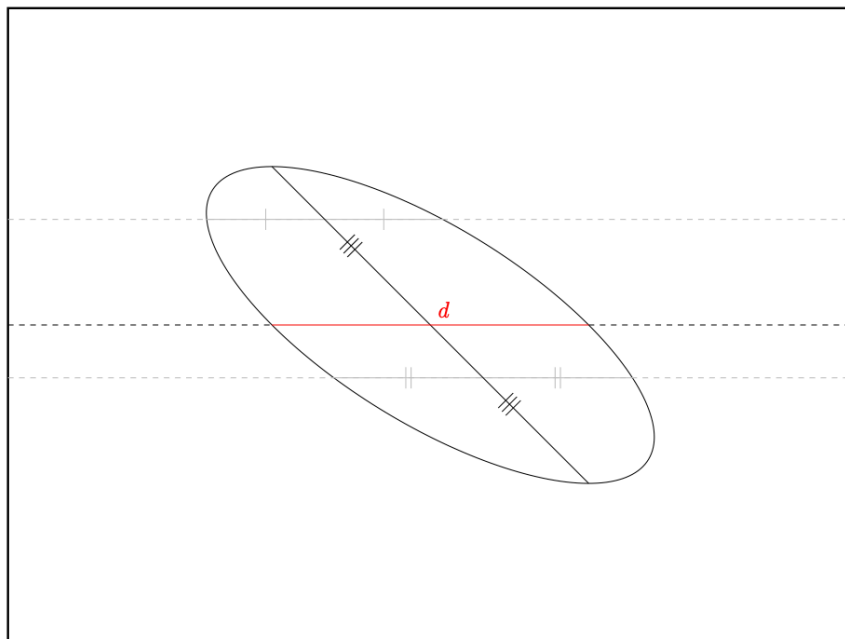
Now we get a segment of length equal to the diameter. To do this, it is enough to draw a horizontal segment through the center of the image $X = 0$.

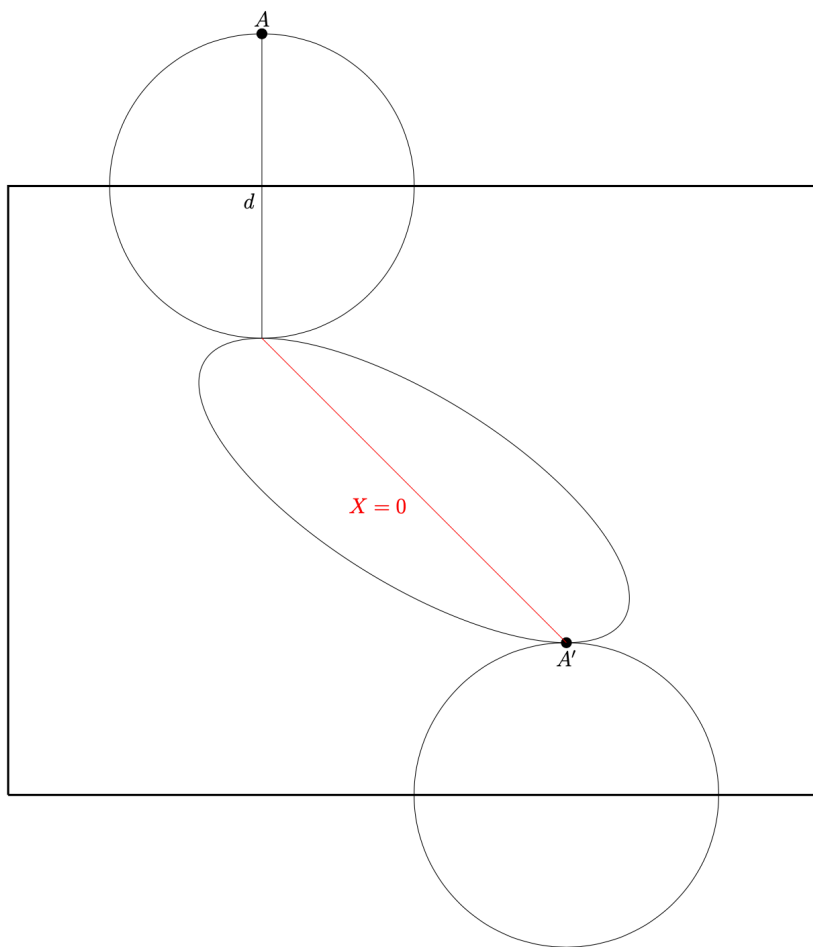
Then there are two options for how photography can take place depending on the speed of the puck. If $u > v \sin \alpha$, then the top point of the image is the top point of the washer. And accordingly, vice versa, if $u < v \sin \alpha$, then the top point of the image is the bottom point of the washer. Let's draw the initial and final states of the puck depending on the speed ratio. $u > v \sin \alpha$:

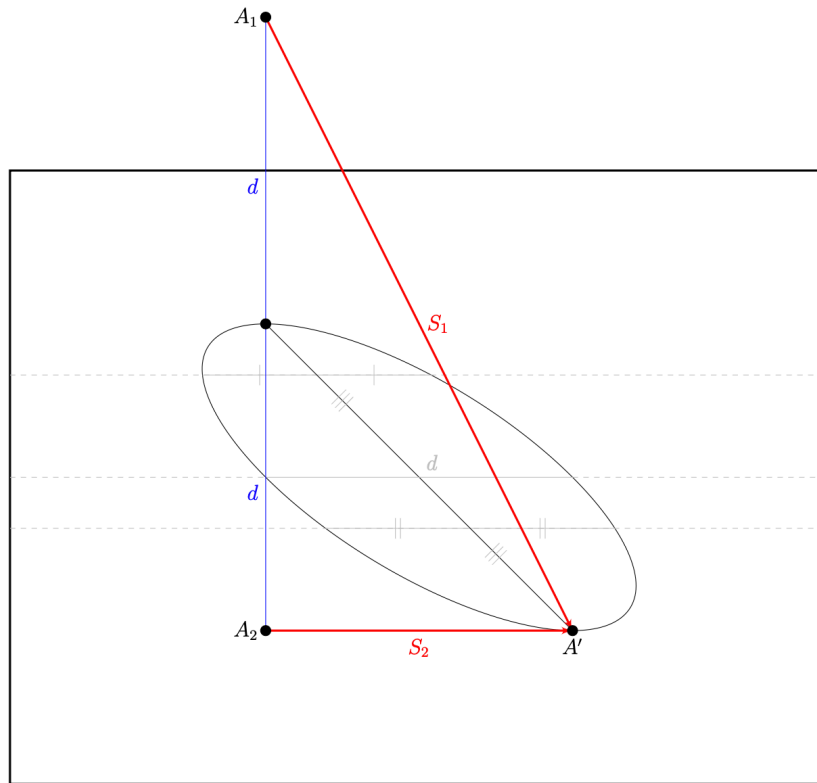
$$u < v \sin \alpha :$$

Note that in this way we can obtain the possible direction of the speed: we obtain the point A by constructing a vertical segment of length d with one end at the top point of the image of the puck, the speed will be directed along $\overrightarrow{AA'}$.









B2

1.00 pts

Using an additional ruler with divisions, find the speed modulus of the puck.

It remains to determine the velocity module. We plotted the displacement of point A during the time of shooting the puck τ . We can express this time in terms of the total photographing time T .

$$\tau = \frac{l}{L}T = 0.5 \cdot T = 5 \text{ ms}$$

By photograph can determine displacement puck:

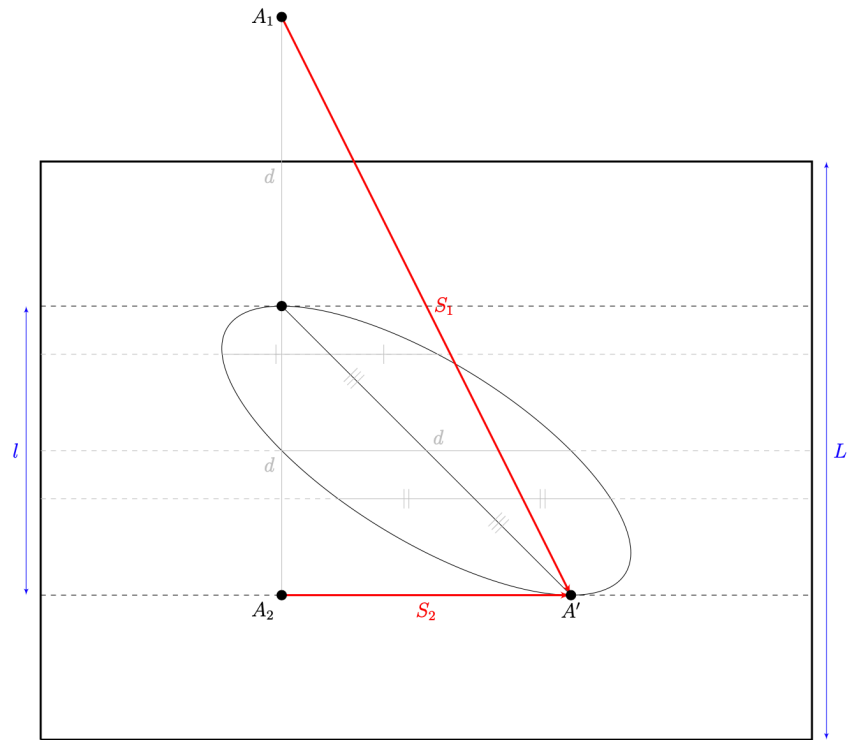
$$\frac{S_1}{d} = 2.24; \quad \frac{S_2}{d} = 1.00.$$

$$S_1 = 17.1 \text{ cm}; \quad S_2 = 7.62 \text{ cm}$$

Remain calculate quantity velocity $v = S/\tau$.

Answer: Answer:

$$v_1 = 34.1 \text{ m/s}; \quad v_2 = 15.2 \text{ m/s}$$



C1

0.50 pts

Show that the linear magnification of a pinhole camera is constant and find α .

Let's consider how an image of a single point is formed.

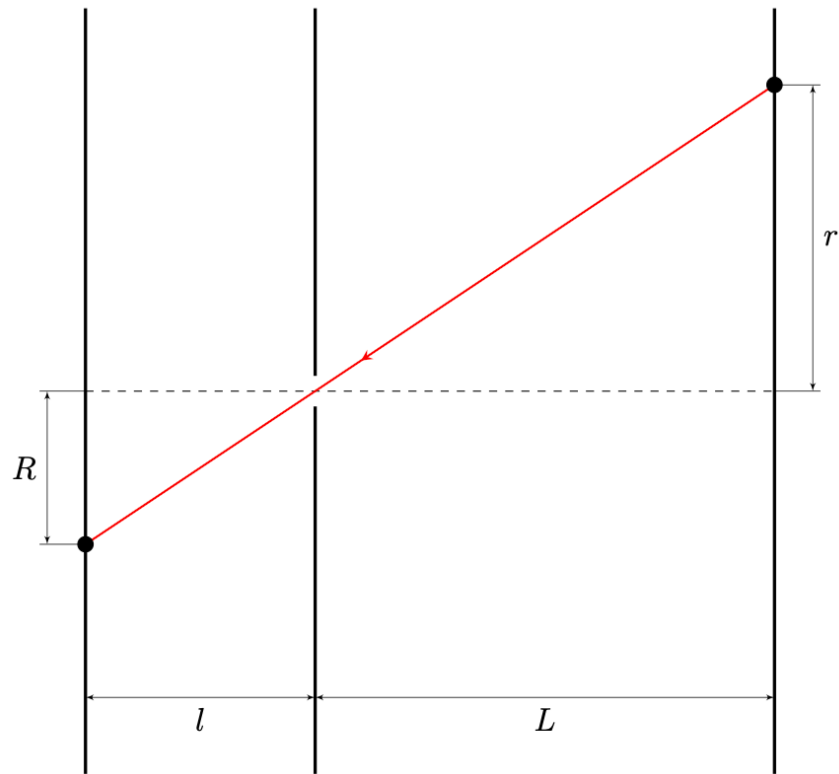
From the similarity we obtain:

$$\frac{R}{r} = \frac{l}{L}.$$

The resulting expression does not depend on the position of the point, so distortion is not observed.

Answer: Answer:

$$\alpha = \frac{l}{L}$$



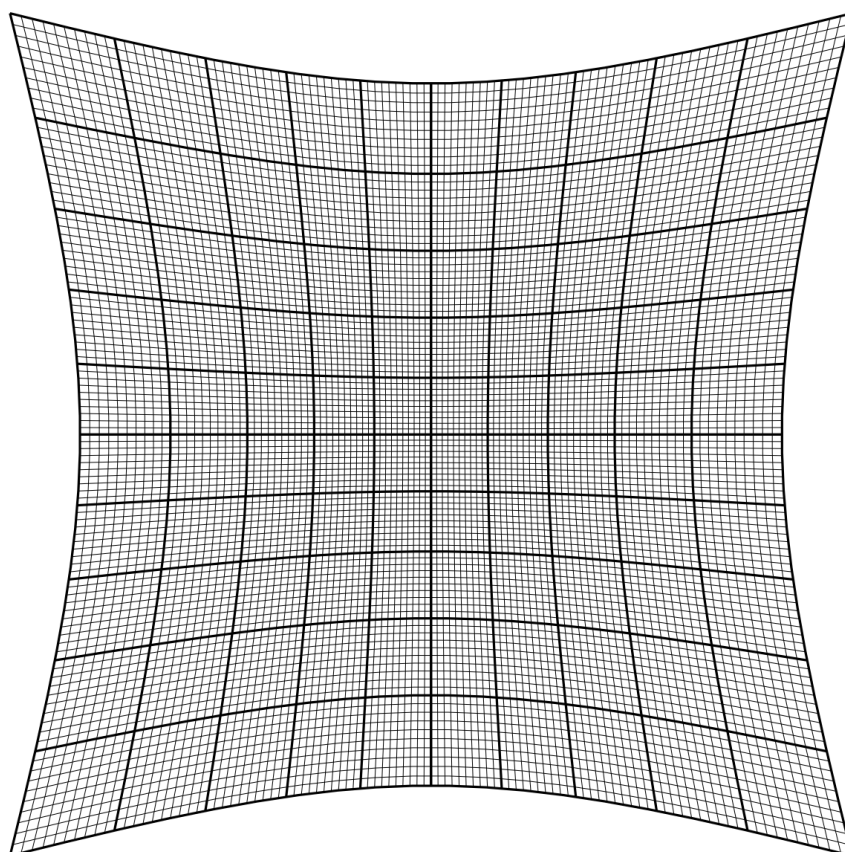
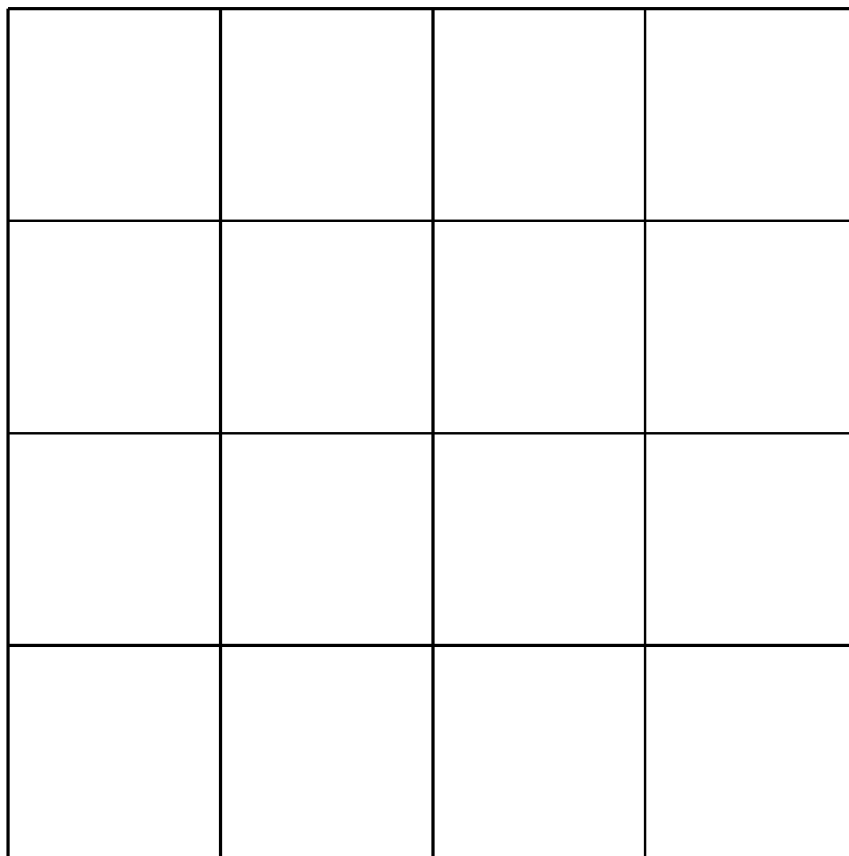
C2

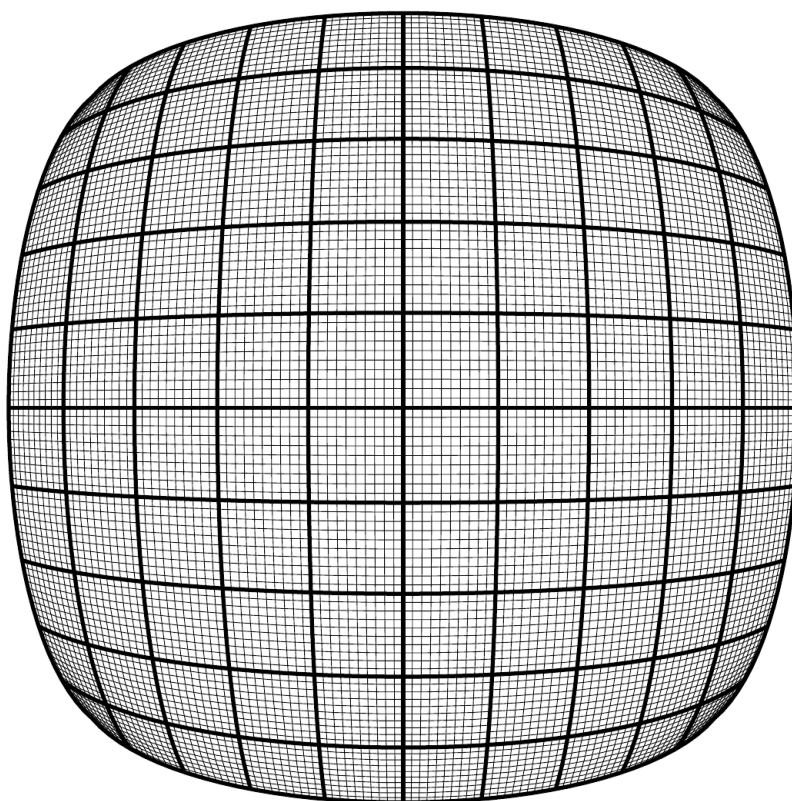
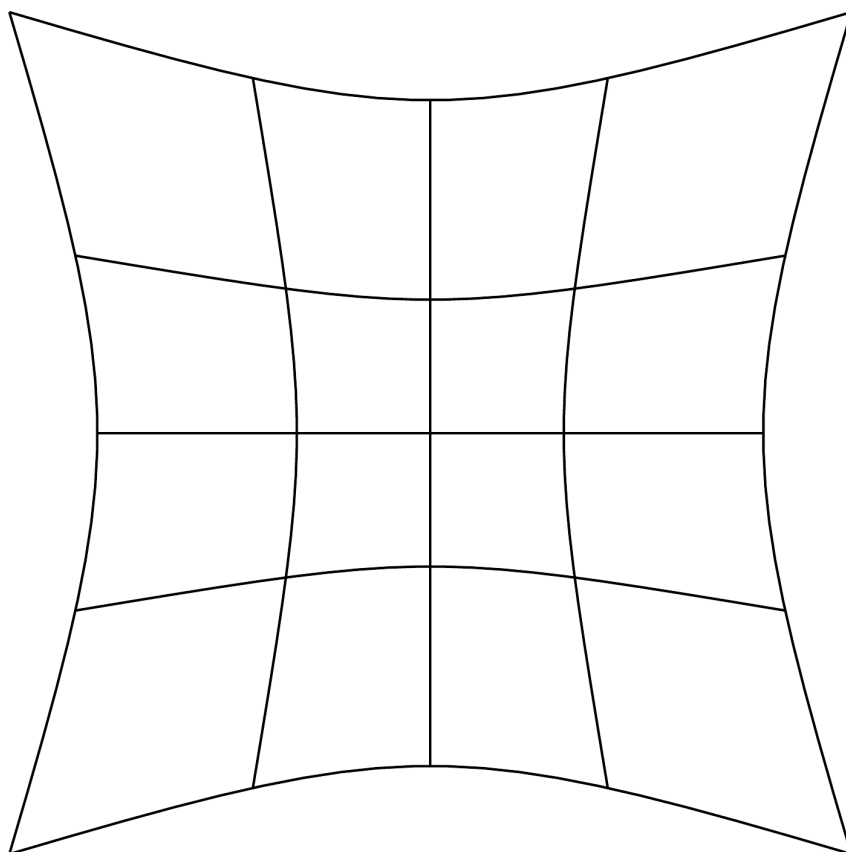
0.50 pts

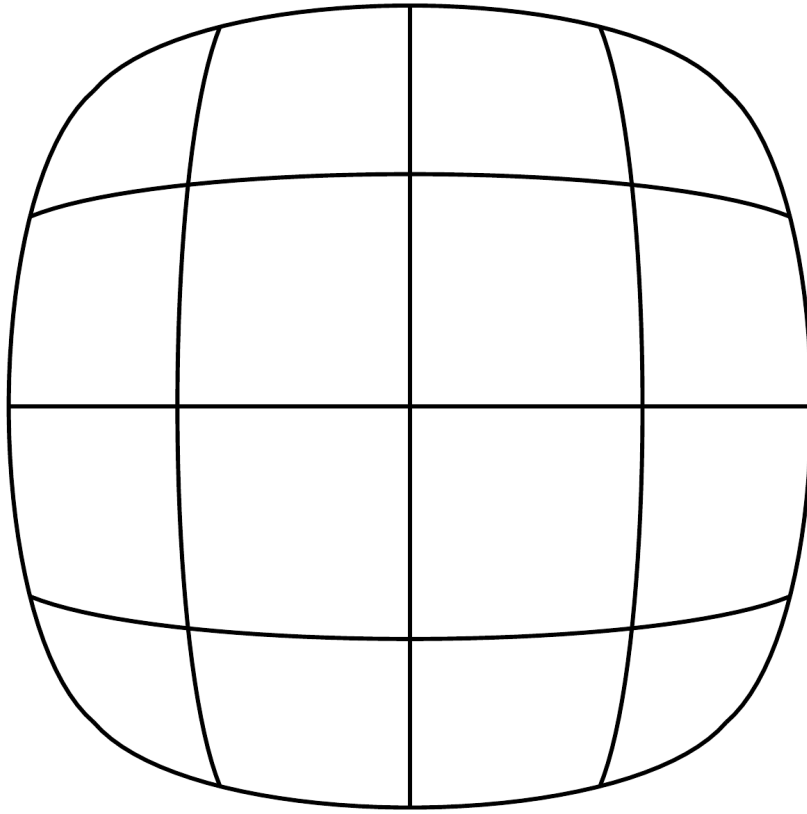
Draw a high-quality view of the photograph of part of the notebook sheet shown in the figure below, in cases $\beta > 0$ (positive distortion) and $\beta < 0$ (negative distortion). Consider that $\alpha > 0$.

In the case of $\beta > 0$ the image of the graph paper (we present it for a more detailed illustration) and the square will look like:

In case of $\beta < 0$:







C3

2.00 pts

Determine the coefficients α and β in this case. What kind of distortion is observed in such a system: positive (α and β of the same sign) or negative (α and β of different signs)?

Note that we are considering third-order effects for paraxial rays, so all formulas can be expanded to third order. Let's find the connection between R and r .

Let's express the dimensions through the angles:

$$R = l \tan \varphi;$$

$$r = (L - h) \tan \varphi + h \tan \psi.$$

Relation angle:

$$n \sin \psi = \sin \varphi.$$

$$\psi = \arcsin \left(\frac{\sin \varphi}{n} \right) \approx \arcsin \left(\frac{\varphi}{n} - \frac{\varphi^3}{6n} \right) \approx \frac{\varphi}{n} - \frac{\varphi^3}{6n} + \frac{\varphi^3}{6n^3} = \frac{\varphi}{n} - \frac{\varphi^3 (n^2 - 1)}{6n^3}$$

$$\tan \psi \approx \psi + \frac{\psi^3}{3} \approx \frac{\varphi}{n} - \frac{\varphi^3 (n^2 - 1)}{6n^3} + \frac{\varphi^3}{3n^3} = \frac{\varphi}{n} - \frac{\varphi^3 (n^2 - 3)}{6n^3}$$

$$\varphi = \arctan \left(\frac{R}{l} \right) \approx \frac{R}{l} - \frac{R^3}{3l^3}$$

$$\tan \psi \approx \frac{R}{nl} - \frac{R^3}{3l^3 n} - \frac{R^3 (n^2 - 3)}{6l^3 n^3} = \frac{R}{nl} - \frac{R^3 (n^2 - 1)}{2l^3 n^3}$$

$$r = \frac{L - h}{l} R + \frac{h}{nl} R - \frac{h (n^2 - 1)}{2l^3 n^3} R^3 = \frac{nL - (n - 1)h}{nl} R - \frac{h (n^2 - 1)}{2l^3 n^3} R^3$$

$$\square 9 \square R. : A = \frac{nL - (n - 1)h}{nl}; \quad B = \frac{h(n^2 - 1)}{2l^3 n^3}.$$

$$r = AR - BR^3$$

$$r \approx A(\alpha r + \beta r^3) - B\alpha^3 r^3 = A\alpha r + (A\beta - B\alpha^3)r^3 : \begin{cases} A\alpha = 1; \\ A\beta - B\alpha^3 = 0. \end{cases}$$

$$\alpha = \frac{1}{A} = \frac{nl}{nL - (n-1)h}$$

$$\beta = \frac{B}{A^4} = \frac{hnl(n^2-1)}{2(nL - (n-1)h)^4}$$

Answer: Answer: Positive distortion is observed.

$$\alpha = \frac{nl}{nL - (n-1)h} \quad \beta = \frac{hnl(n^2-1)}{2(nL - (n-1)h)^4}$$

